Experimental Analysis of Public Policies

Session 7

Regression Discontinuity

Etienne Dagorn
University of Lille

Session Objectives

By the end of this session, students should be able to:

- audit an institutional rule by naming the running variable, cutoff, and treatment margin;
- read an RDD graph before reading a coefficient table;
- define the causal cutoff estimand before the observed discontinuity;
- state continuity and no-manipulation logic in potential-outcomes language;
- distinguish sharp RDD from fuzzy RDD;
- interpret bandwidth, local fitting, diagnostics, and inference as design-supporting choices;
- explain why RDD estimates are local and how fuzzy RDD reconnects to IV and LATE.

Roadmap

Institutional Rule and Local Comparison

Graph-First Intuition

Sharp RDD: Assumptions and Estimation

Diagnostics and Threats

Fuzzy RDD and External Validity

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From Time Designs to Thresholds

- DiD and event studies used timing and comparison groups to build counterfactual trends.
- IV identified a local effect for people moved by an instrument.
- RDD is local in a different way: units just around an institutional cutoff.

> **Central question**
>
> Can an institutional threshold create a locally credible comparison?

Course link: we again ask where the identifying variation comes from, which assumption makes it causal, and which population the estimate describes.

The RDD Idea

- A rule changes treatment, eligibility, or treatment probability when a score crosses a cutoff.
- Units very close to the cutoff should be similar in the running variable.
- Under continuity and no other discontinuous change at the cutoff, an outcome jump can be interpreted as causal.
- The estimand is local to the cutoff, not an average over the full score distribution.

In plain words: RDD compares just-missing and just-qualifying units, not all treated and untreated units.

RDD Workshop: Audit the Rule

Scenario: pupils scoring at least 70/100 receive an automatic scholarship.

1. What is the running variable?
2. What is the cutoff?
3. Is the design sharp or fuzzy?
4. Which pupils define the local comparison?
5. What manipulation or retesting behaviour would worry you?

> **Debrief**
>
> The credibility of RDD comes from the institutional rule and local comparability, not from fitting a flexible curve.

Expected output: one paragraph naming the running variable, cutoff, local population, and manipulation risk.

Threshold Designs in Practice

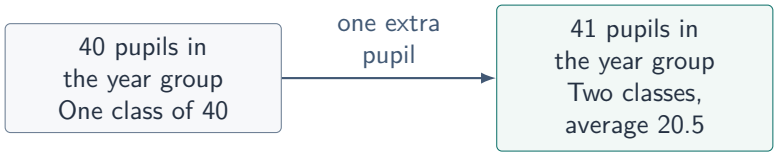
Institutional rule	Running variable	Possible outcome
Scholarship eligibility	test score	enrolment or graduation
Age eligibility	age	employment or health
Election victory	vote share	policy spending
Class-size rule	cohort enrolment	achievement
Income threshold	reported income	benefit take-up

Historical anchor

Maimonides' Rule works pedagogically because the institution creates a visible threshold in class size.

Angrist and Lavy: Can Class Size Improve Achievement?

Problem: small classes may be allocated to pupils already facing difficulties. **Institution:** Israeli primary schools use Maimonides' rule, with a class-size cap of 40.

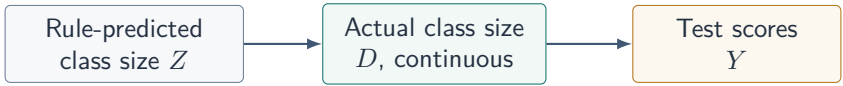


Predict before reading the result

If smaller classes improve learning, what pattern should appear in test scores when enrolment crosses this rule?

Angrist and Lavy (1999), QJE, pp. 535 and 540–542. Rule schema, not observed class sizes.

What the Class-Size Paper Teaches About IV and Locality



- The authors use rule-induced variation to instrument actual class size.
- They find gains from smaller classes for fourth and fifth graders, with little evidence for third graders.
- **Interpretation:** this is a continuous-treatment IV application; it is not literally the binary-scholarship complier example.

Ask: does the enrolment rule change other educational inputs or pupil composition?

Angrist and Lavy (1999), QJE, pp. 533–536.

Running Variable and Cutoff

$$X_i = \text{running variable}, \qquad c = \text{cutoff}.$$

Sharp rule

$$D_i = 1[X_i \geq c].$$

- In this session, X_i denotes the running variable; other covariates can be labelled W_i.
- In a sharp RDD, the rule determines treatment received.
- In a fuzzy RDD, the cutoff changes $Pr(D_i = 1 \mid X_i)$ rather than deterministically assigning everyone.
- The comparison is local around c.

Roadmap

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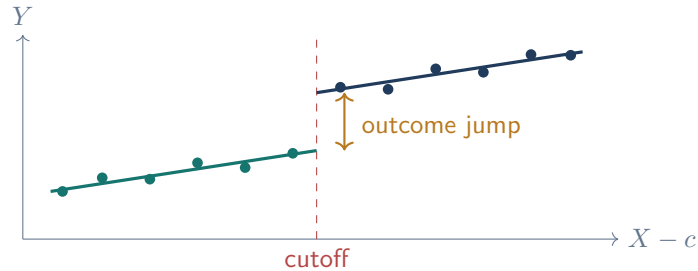
Graph-First Intuition

Sharp RDD: Assumptions and Estimation

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Fuzzy RDD and External Validity

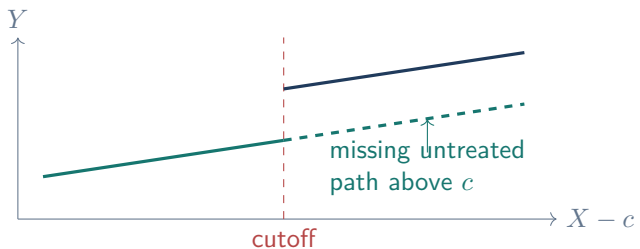
Read the Jump Before Reading a Regression Table



The running variable can predict outcomes smoothly. The design uses the **discontinuity at the rule**.

Illustrative schema; not estimates from a paper.

Draw the Missing Counterfactual



Illustrative schema; not estimates from a paper.

Just below c, we observe pupils without the scholarship. Can their outcome path approximate what just-qualifying pupils would have experienced **without it**?

Causal Cutoff Estimand

$$\tau_{SRD}(c) = E[Y_i(1) - Y_i(0) \mid X_i = c].$$

- This is the sharp-RDD causal target: the treatment effect at the cutoff.
- With a continuous running variable, read $X_i = c$ as the limiting population around the cutoff.
- It is not the ATE for all eligible units and not the ATT for all treated units.
- Define this causal object before writing the observed jump.

Policy interpretation: the estimand may inform a small threshold change if effects and behaviour remain comparable.

Observed Discontinuity and Identification

$$\Delta_Y(c) = \lim_{x \downarrow c} E[Y_i \mid X_i = x] - \lim_{x \uparrow c} E[Y_i \mid X_i = x].$$

Identification statement

Under the sharp rule and continuity of potential outcomes at c,

$$\Delta_Y(c) = \tau_{SRD}(c).$$

- $\Delta_Y(c)$ is the observed discontinuity.
- $\tau_{SRD}(c)$ is the causal estimand.
- The assumption links the observed jump to the missing counterfactual.

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Continuity Assumption

Continuity

In each treatment state, the expected potential outcome is continuous at the cutoff:

$$\lim_{x \downarrow c} E[Y_i(d) \mid X_i = x] = \lim_{x \uparrow c} E[Y_i(d) \mid X_i = x], \quad d \in \{0, 1\}.$$

- Units just below and above c should not differ discontinuously for other reasons.
- The running variable may affect outcomes; it should not create a separate jump at c absent treatment.
- Diagnostics can support this story, but they cannot prove the counterfactual smoothness.

Why Continuity Turns a Jump into an Effect

The sharp rule tells us which potential outcome is observed on each side:

$$\begin{aligned}\Delta_Y(c) &= \lim_{x \downarrow c} E[Y_i(1) \mid X_i = x] - \lim_{x \uparrow c} E[Y_i(0) \mid X_i = x] \\ &= E[Y_i(1) \mid X_i = c] - E[Y_i(0) \mid X_i = c] \\ &= \tau_{SRD}(c).\end{aligned}$$

- First line: assignment and consistency.
- Second line: continuity of both conditional potential-outcome means.
- The fitted jump estimates these limits using observations near c.

Check: which step fails if another grant begins at exactly the same score?

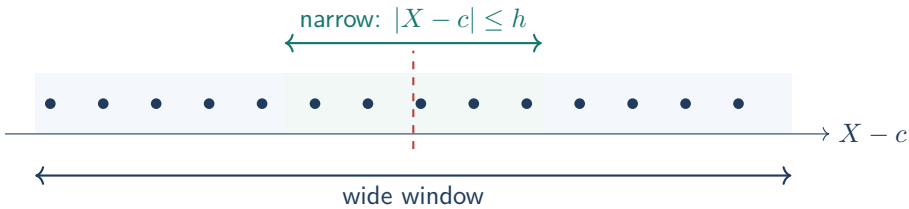
No Other Jump at the Rule

- The cutoff should not bundle multiple policies unless the estimand is the bundle.
- Administrative appeals, retesting, or reclassification can create discontinuous selection.
- A simultaneous rule for another programme can contaminate the outcome jump.

Causal language

Say “the jump is interpreted as causal under continuity and no other discontinuous change”, not simply “the jump is the effect”.

Bandwidth: How Local Should the Comparison Be?



Narrow: closer comparisons, fewer observations, less precision.

Wide: more observations, but greater risk from fitting the wrong curve.

A Local Estimate Has Both Noise and Approximation Bias

$$\widehat{\tau}(h) - \tau_{SRD}(c) \;=\; \text{local approximation bias}(h) + \text{sampling noise}.$$

- Wider windows can reduce noise while increasing curvature bias.
- A conventional OLS interval may ignore bias that matters for coverage.
- Report bandwidth, kernel, observations on each side and robust bias-corrected intervals.

The target stays local

Changing h usually changes the approximation and precision, not the intended cutoff estimand. Do not choose the window for statistical significance.

Cattaneo, Idrobo and Titiunik (2020), Foundations, estimation and inference.

The Equation Behind Two Local Lines

Keep observations with $|X_i - c| \leq h$, then fit:

$$Y_i = \alpha + \tau D_i + \beta_1(X_i - c) + \beta_2 D_i(X_i - c) + u_i.$$

Just below the cutoff

$$\widehat{Y} = \widehat{\alpha} + \widehat{\beta}_1(X - c)$$

Just above the cutoff

$$\widehat{Y} = (\widehat{\alpha} + \widehat{\tau}) + (\widehat{\beta}_1 + \widehat{\beta}_2)(X - c)$$

Interpret the coefficient

At $X = c$, the estimated vertical gap is $\widehat{\tau}$. Allow separate slopes; avoid high-order global polynomials.

Polynomial Warning

Avoid overfitting

High-order global polynomials can create artificial jumps and unstable estimates.

- Prefer local methods.
- Show sensitivity to bandwidth and functional form.
- Let the graph and institutional rule guide interpretation.

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RDD Diagnostic Workshop

Check	What would worry us?	What it can show
Density	bunching at c	possible manipulation
Covariates	jumps in W_i	comparability concern
Placebo cutoffs	jumps where no rule exists	functional-form concern
Bandwidths	sign changes under reasonable windows	fragile estimation
Institutional audit	another rule at c	bundled treatment

RDD Diagnostic Workshop: Activity

Visible output

For the scholarship rule, choose the two most important diagnostics and explain which assumption each one probes.

- Name the diagnostic.
- State the threat it probes.
- Explain why the diagnostic cannot prove the identifying assumption.

Covariate Continuity

- Pre-treatment covariates should not jump at the cutoff.
- Check age, baseline outcomes, demographics, prior trends, or institutional variables.
- Covariate discontinuities suggest the cutoff captures more than treatment.
- Smooth covariates support the design but do not prove continuity of potential outcomes.

Example: if baseline test scores jump exactly at a scholarship cutoff beyond the mechanical threshold variable, another selection process may be operating.

Manipulation and Density

- Units may manipulate the running variable to cross the threshold.
- Schools may reclassify pupils; firms may adjust reported employment; households may time eligibility.
- A density jump at c is most worrying when agents know the cutoff and can precisely affect X_i.
- A smooth density supports, but does not prove, local comparability.

Threat

Precise sorting can change the composition on either side and undermine continuity of potential outcomes.

Placebo Cutoffs

- Estimate discontinuities at thresholds where no treatment changes.
- Large placebo jumps weaken credibility.
- Placebo cutoffs should be chosen transparently.
- They help detect functional-form artefacts or unstable relationships between X_i and Y_i.

Good placebo: choose fake cutoffs on both sides of the true cutoff, while keeping enough observations nearby.

Donut RDD

- Exclude observations extremely close to the cutoff.
- Useful for local heaping, appeals, retesting, or administrative quirks around c.
- It removes nearby support and requires extrapolation back to the intended cutoff target.
- If manipulation changes who appears on each side, a donut check does not automatically repair the design.

Example: if scores can be appealed within one point of the cutoff, compare results with those observations removed, but still discuss selection.

Threats

Threat	Diagnostic or response
Manipulation	density and institutional checks
Other policy cutoff	rule audit and covariate continuity
Functional form	graph, bandwidth, and specification sensitivity
Noisy local estimate	confidence intervals and transparent bandwidth choice
Local external validity	describe units near cutoff

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Fuzzy RDD

$$Z_i = 1[X_i \geq c], \qquad D_i \neq Z_i \text{ for some units.}$$

- Treatment probability jumps at the cutoff but not from 0 to 1.
- The cutoff indicator Z_i becomes a candidate instrument for treatment received D_i.
- The estimand is a local IV effect for compliers at the cutoff.
- A first-stage jump is necessary, but not enough for validity.

Connection: fuzzy RDD imports the IV logic from Session 4 into a threshold setting.

A Local Wald Ratio: The Same Logic as Session 4

For any variable V, let $\Delta_V(c)$ be its jump from just below to just above c. The local Wald ratio identifies the target **under the local IV assumptions**:

$$\underbrace{\tau_{FRD}(c)}_{\text{local complier effect}} = \frac{\overbrace{\Delta_Y(c)}^{\text{outcome jump}}}{\underbrace{\Delta_D(c)}_{\text{receipt jump}}}.$$

Two-minute calculation: illustrative scholarship numbers

Enrolment jumps by 6 percentage points; receipt jumps by 60 percentage points. Calculate the ratio. Whose effect is it, under the assumptions?

$$\widehat{\tau}_{FRD} = \frac{0.06}{0.60} = 0.10 \quad \Rightarrow \quad \text{10 pp for } \textbf{cutoff compliers}.$$

Fuzzy RDD Assumptions at the Cutoff

Assumption	Local question
Relevance	Does crossing c visibly change D_i?
Continuity	Would potential outcomes and potential treatment behaviour be smooth through c?
Exclusion	Does the cutoff affect Y_i only through treatment received?
Monotonicity	Is anyone less likely to receive treatment because they are just above c?

Same IV discipline

Do not treat a threshold first stage as proof of exclusion or monotonicity.

Why the Ratio Describes Cutoff Compliers

Let $D_i(z)$ be receipt if eligibility were z. For binary receipt, define

$$C_i = 1\{D_i(1) > D_i(0)\}, \qquad D_i(1) \geq D_i(0).$$

Under local continuity, exclusion, monotonicity and a positive first stage:

$$\Delta_D(c) = \Pr(C_i = 1 \mid X_i = c),$$
$$\Delta_Y(c) = \Pr(C_i = 1 \mid X_i = c) \underbrace{E[Y_i(1) - Y_i(0) \mid C_i = 1, X_i = c]}_{\tau_{FRD}(c)}.$$

The complier share cancels. Always-takers and never-takers do not change receipt.

A small denominator is an inference problem

A weak jump can make the ratio unstable; report both jumps and their uncertainty. A strong first stage still does not establish exclusion.

External Validity

- Sharp RDD estimates effects for units near the cutoff.
- Fuzzy RDD estimates effects for local compliers near the cutoff.
- Effects may differ for units far above or below the threshold.
- This local estimate can be highly policy-relevant when the decision is whether to move the cutoff.

Policy reflex: distinguish moving a threshold from universal expansion; they often require different estimands.

Inference and Reporting

- Use standard errors compatible with local estimation and bandwidth choice.
- Account for clustering if treatment or shocks vary by group.
- Report confidence intervals, not only point estimates.
- For fuzzy designs, report the treatment jump as carefully as the outcome jump.

Minimum reporting: graph, rule, bandwidth, observations on each side, main estimate, sensitivity checks, density test, and covariate checks.

R Mini-Lab: Does the Local Estimate Depend on the Window?

Simulated outcome-score data: cutoff centred at zero.

```
source("r_exercises/week_07_rdd.R")
h <- c(10, 20, 35)
data.frame(bandwidth = h,
  jump = sapply(h, function(b) estimate_jump("Y", b)))
estimate_jump("W", 20)  # baseline covariate diagnostic
```

Eight minutes: compute and interpret

Does the sign or magnitude change? Is the jump in W small? Write one sentence about the local effect and one about what this diagnostic cannot establish.

These are point estimates; a full empirical analysis also needs appropriate RDD intervals.

Exercise Bridge

- Theory sheet: `homework/session_07_homework.md`.
- QCM: `assessments/qcm/session_07_qcm.md`.
- Applied exercise: assess an eligibility-threshold design.
- Exam skill: explain the local estimand and validity checks.

Exercise rhythm: identify the rule, state continuity, inspect diagnostics, then interpret the local effect.

Design Checklist

1. What is the running variable?
2. What is the cutoff rule?
3. Is treatment sharp or fuzzy?
4. What is the causal cutoff estimand?
5. What assumption links the observed jump to the causal effect?
6. Is manipulation plausible?
7. Do covariates or other rules jump at the cutoff?
8. How local is the estimand?

Transition to Session 8

- RDD is powerful when a rule creates local variation.
- Its credibility is often local: cutoff effects or cutoff-complier effects.
- Some policies affect one city, region, or country with no lottery, no clean threshold, and few treated units.
- Session 8 asks how to construct a counterfactual, assess heterogeneity and external validity, and stress-test credibility.

Bridge question: when there is no threshold and no many-treated-units panel, what comparison can we defend?

Key Takeaways

- RDD starts from an institutional rule, not from curve fitting.
- The causal cutoff estimand is distinct from the observed discontinuity.
- Continuity of potential-outcome means carries identification; sorting and other cutoff rules can threaten it.
- Bandwidth choice trades local credibility against precision.
- Fuzzy RDD is a local IV design with IV assumptions at the cutoff.

One-sentence memory aid: RDD is a local comparison made credible by a discontinuous rule and smooth counterfactual outcomes.

Appendix: Fuzzy RDD as Local 2SLS

Within a window around c, define $R_i = X_i - c$, $Z_i = \mathbf{1}\{R_i \geq 0\}$. An illustrative local-linear specification is:

$$D_i = \pi_0 + \pi_1 Z_i + \pi_2 R_i + \pi_3 Z_i R_i + \eta_i,$$

$$Y_i = \alpha + \tau \widehat{D}_i + \beta_1 R_i + \beta_2 Z_i R_i + \varepsilon_i.$$

Instrument D with Z, including R and ZR as the same controls in both stages. With the same window and weights, the just-identified coefficient is the ratio of fitted outcome and receipt jumps.

Implementation

Use an IV/RDD command for appropriate inference; naive second-stage OLS standard errors are incorrect.

Appendix: What If Scores Are Discrete or Missing Near the Cutoff?

- A continuity argument uses outcomes approaching c from both sides.
- With a few distinct scores, more pupils at each score do not fill the gaps.
- A donut exclusion also leaves a hole that must be extrapolated across.
- State the extra smoothness or local-randomisation assumptions needed.

Interpretation check

If only scores 60, 70 and 80 are possible, does a million observations make the comparison infinitesimally local?

Appendix: Extensions

- Regression kink designs use changes in slope rather than jumps.
- Geographic RDD uses borders as cutoffs.
- Multiple cutoffs can improve external-validity discussion.
- These extensions keep the same local-comparison logic.

Student focus: these extensions are optional; the core exam skill is the sharp/fuzzy cutoff design.

References: Threshold and IV Applications

- Lee, D. S. (2008). "Randomized Experiments from Non-random Selection in U.S. House Elections." *Journal of Econometrics*, 142(2), 675–697.
  Author-hosted article.
- Angrist, J. D., and Lavy, V. (1999). "Using Maimonides' Rule to Estimate the Effect of Class Size on Scholastic Achievement." *QJE*, 114(2), 533–575.
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- Cattaneo, M. D., Idrobo, N., and Titiunik, R. (2020). *A Practical Introduction to Regression Discontinuity Designs: Foundations*.